Generating Chest X-Rays with a Variational Autoencoder

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Abstract

This report presents a variational autoencoder (VAE), implemented in PyTorch and trained on the 1,341 normal chest radiographs in the training split of the Kaggle chest X-ray pneumonia dataset. The encoder and decoder are fully connected networks, images are 128×128 grayscale, and the latent code has 256 dimensions. The report covers the hyperparameters, training loss curves, the Fréchet Inception Distance (FID) and Inception Score (IS) from pytorch-ignite, and generated images. The samples show the overall anatomy of a chest X-ray and vary from sample to sample, but they are blurrier than real images and lose fine texture. The code is at `https://github.com/Supernova1114/ece-285-hw1-vae`.

1 Introduction

A VAE pairs a decoder $p_\theta(x \mid z)$ with an encoder $q_\phi(z \mid x)$ and trains both by maximizing the evidence lower bound (ELBO) under the prior $p(z) = \mathcal{N}(0, I)$. Decoding $z \sim \mathcal{N}(0, I)$ then generates new images. Here this is used to generate normal chest X-rays, which are scored with FID and IS.

2 Method

Data. The data is the chest X-ray pneumonia dataset from Kaggle. The model trains only on the 1,341 images in `train/NORMAL`; the 234 images in `test/NORMAL` serve only as an FID reference. Each image is converted to grayscale, resized so its shorter side is 128 pixels, center-cropped to 128×128 and scaled to $[0, 1]$. No augmentation is used.

Architecture. The encoder flattens the image into $128^2 = 16{,}384$ pixels and applies two fully connected layers of width 1,024 with ReLU. Two linear heads output the mean $\mu(x)$ and log variance $\log \sigma^2(x)$ of $q_\phi(z \mid x)$, both in $\mathbb{R}^{256}$. The reparameterization trick, $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, lets gradients flow through sampling. The decoder mirrors the encoder ($256 \to 1{,}024 \to 1{,}024 \to 16{,}384$) and ends in a sigmoid so pixels lie in $[0, 1]$.

Loss. The loss for each image has two terms:

$$\mathcal{L}(x) = \|x - \hat{x}\|^2 + \beta \, \mathrm{KL}\big(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I)\big). \tag{1}$$

The first is the squared error between the image x and the decoder's output $\hat{x}$, summed over all pixels. The second is the KL divergence, which measures how far the encoder's distribution for z is from a standard Gaussian. Keeping it small is what lets random $z \sim \mathcal{N}(0, I)$ decode into realistic images. $\beta = 1$ is the weight on the KL term.

Training and evaluation. Table 1 lists the settings. Training uses Adam, and the final weights are kept. FID and IS use pytorch-ignite's default `FID` and `InceptionScore` metrics, which run an

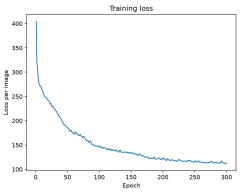
Table 1: Hyperparameter settings.

Setting	Value
Image size	128×128, grayscale
Training images	1,341 (`train/NORMAL`)
Latent dimension	256
Hidden layers	$2 \times 1{,}024$ each, ReLU
Parameters	36.5M
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	10^{-3}, constant
Batch size / epochs	64 / 300
KL weight β	1.0

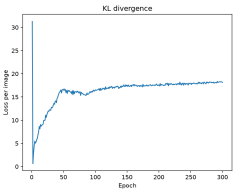
Table 2: FID and IS from pytorch-ignite. FID is measured against `train/NORMAL`.

Images	FID ↓	IS ↑
Real X-rays*	0.024	2.03
VAE samples	0.180	1.74
Untrained VAE	0.203	1.02

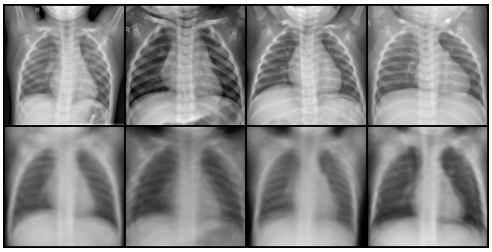
*FID: 234 `test/NORMAL` images against 234 training images. IS: all 1,341 training images.



(a) Total loss



(b) KL divergence



(c) Real (top) and generated (bottom)

Figure 1: (a, b) Training loss and KL divergence per image, averaged over each epoch. (c) Four real training images above four images generated by decoding $z \sim \mathcal{N}(0, I)$.

ImageNet-pretrained InceptionV3 on images upsampled to 299×299 with the gray channel copied into three color channels. 1,341 generated images are compared with the 1,341 training images. For scale, real images and an untrained model are scored too.

3 Results

Training curves. Figure 1(a, b) shows the loss and KL divergence per image. The loss drops quickly at first and levels off near 112 by epoch 300. The KL divergence settles near 18 rather than falling to zero, so the decoder makes use of z.

FID and IS. In Table 2, lower FID and higher IS are better. Real test images score an FID of 0.024 against the training images, roughly the best a model can reach, while an untrained model that outputs noise scores 0.203. The VAE's 0.180 closes only 13% of that gap, consistent with its blur. Its IS of 1.74 is near the real images' 2.03.

Generated images. The samples in Figure 1(c) are recognizable frontal chest X-rays, with two dark lung fields, the spine, curved ribs and the diaphragm, and they differ in brightness, contrast and body size. Next to the real images they are clearly blurrier: rib edges are soft and the fine lung texture is missing.